Xiaojian Xu

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Foundation AI Team ⋄ GE HealthCare

About me

Research

- My current research focuses on training and developing large language models (LLMs), vision-language models (VLMs), and multimodal foundation models across the model lifecycle, including pretraining, fine-tuning, and post-training. My work draws on natural language processing, computer vision, and computational imaging to advance learning methods and address challenging problems across domains.

Expertise

- Computational imaging/sensing, Image reconstruction, Multimodal models, Vision-Language models, Foundation models, Transformers, Diffusion models, Generative AI, Computer vision, Large-scale optimization, Image analysis, Signal/Image processing, Neural rendering, Inverse problems

Education

Washington University in St. Louis (WashU), USA *08/2017–08/2022*

- Ph.D. in Computer Science (GPA: 3.87/4.00)

University of Electronic Science and Technology of China (UESTC), China *09/2014–06/2017*

- M.E. in Communication & Information Engineering (Graduated with honors)

University of Electronic Science and Technology of China (UESTC), China *09/2010–06/2014*

- B.E. in Communication Engineering (GPA: 3.89/4.00)

Work Experience

GE HealthCare (GEHC) *05/2024 – present*
AI Research Scientist *Bellevue/Seattle*

- Worked on pretraining foundation models for healthcare applications, as well as supervised fine-tuning (SFT) and post-training of vision-language models (VLMs).
- Developed multimodal models integrating language and medical imaging data. Representative publications: arXiv 2026 [p 1], npj Digital Medicine 2026 [j 1], arXiv 2025 [p 2], and KDD 2025 [c 1].
- Led and contributed to developing AI-based healthcare workflow automation systems, with applications such as automated image acquisition, medical image analysis, information retrieval, and report generation, etc.

University of Michigan (UMich) *08/2022 – 05/2024*
Postdoctoral Research Fellow *Ann Arbor*

- Developed an efficient 3D deep learning network for various imaging applications.
- Designed a state-of-the-art learning-based feature extraction DL network with subpixel level accuracy.
- Designed a powerful deep learning diffusion-model-based image prior for imaging.

Meta Reality Labs Research (MRLR) *05/2021 – 08/2021*
Research Intern *Seattle*

- Programed the end-to-end ray-tracing model for peripheral display system in Oculus AR/VR devices.
- Developed multiple optimization algorithms for solving the display calibration problem in Oculus.
- Designed a novel and efficient eye-movement-based camera-to-display mapping for its real-time rendering and view synthesis using implicit neural representation (NeRF).

Mitsubishi Electric Research Laboratories (MERL) *05/2019 – 08/2019*
Research Intern *Boston*

- Developed two distinct methods, optimization and deep learning method, for solving the 3D tomographic imaging problem, respectively.

Research Experience

Multimodal Foundation AI Models *05/2024 – Present*

- Developed multimodal foundation models and downstream applications for healthcare modalities, including MRI, CT, and X-ray (representative publications: [p 2], [j 1]).
- Advanced vision-language models (VLMs) for healthcare applications through supervised fine-tuning (SFT) and post-training (representative publications: [p 1], [c 1]).

Physics-Informed Machine Learning Algorithms *08/2020 – 05/2024*

- Developed various physics-informed deep learning methods for different imaging tasks. (Representative publication: [c 5]).
- Extensively investigated and developed variants of learning-based optimization algorithms for various imaging tasks by combining the imaging models with the deep-learning priors, in both theory and practice.

Compressed and Stochastic AI Algorithms for Large-Scale Optimization *07/2018 – 05/2024*

- Developed stochastic/compressed variants of optimization- and deep learning algorithms with convergence guarantee.
- Designing memory-efficient networks architectures for 3D imaging applications.

Other Earlier Research Experience *03/2014 – 06/2017*

- Routing and resource scheduling algorithms for large-scale software defined networks (SDN).

Awards & Honors

Honors

- Honored Ph.D student in Computer Science & Engineering Department, WashU *2021*
- Outstanding Graduate Student, UESTC *2017*

Scholarship

- Graduate Student First-Rank Academic Scholarship, UESTC *2016*
- Graduate Student Second-Rank Academic Scholarship, UESTC *2015*
- Graduate Student First-Rank Academic Scholarship, UESTC *2014*
- National Inspirational Scholarship, UESTC *2013*
- People's First-Rank Scholarship, UESTC *2012*
- National Inspirational Scholarship, UESTC *2011*

Invited talks

- Washington State University *03/2024*
- Johns Hopkins University *02/2024*
- SIAM Conference on Imaging Science, Deep Learning for Imaging Science *05/2024*
- Michigan State University *12/2023*
- Los Alamos National Lab *11/2023*
- Stanford University, Laboratory of AI in Medicine and Biomedical Physics *01/2022*
- University of Michigan, Image and Signal Processing Group *12/2021*
- UCLouvain, Image and Signal Processing Group Seminar *09/2020*

Professional Service

- **Committee member**: IEEE Bio Imaging and Signal Processing Technical Committee (BISP TC)
- **Conference reviewer**: Conference on Neural Information Processing Systems (NeurIPS); International Conference on Learning Representations (ICLR); IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); IEEE/CVF International Conference on Computer Vision (ICCV); European Conference on Computer Vision (ECCV); IEEE/CVF Winter Conference on Applications of Computer Vision (WACV); British Machine Vision Conference (BMVC); IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP); IEEE International Symposium on Biomedical Imaging (ISBI).
- **Journal reviewer**: IEEE Transactions on Image Processing (TIP), IEEE Transactions on Computational Imaging (TCI), IEEE Transactions on Medical Imaging (TMI), IEEE Signal Processing Letters (SPL), IEEE Open Journal of Signal Processing (OJSP), Signal, Image and Video Processing, Nuclear Instruments and Methods in Physics Research-A (NIM-A), Applied Mathematical Modelling, Optics Communications, Scientific Reports, Signal Processing

Teaching Experience

(Head) Teaching Assistant

- ESE 415 Optimization, WashU, 2019 Spring, 2021 Spring
- CSE 534A/ESE 513 Large-Scale Optimization, WashU, 2020 Fall

Publications

Preprints *(* indicates equal contribution)*

[p 1] C. Y. Park, J. Mao, **X. Xu**, T. Kass-Hout, U. S. Kamilov, and C. Xiao, "Next-Dense-Stride Prediction for Multimodal Autoregressive Visual Modeling." arXiv:2607.09892, 2026. [Paper]

[p 2] N. Singla, K. Koos, F. Haddadpour, A. H. Shandiz, L. Chum, **X. Xu**, Q. Jin, and E. Bas, "Multi Anatomy X-Ray Foundation Model." arXiv:2509.12146, 2025. [Paper]

Journals *(* indicates equal contribution)*

[j 1] Z. Yang, N. DSouza, I. Megyeri, **X. Xu**, A. H. Shandiz, F. Haddadpour, K. Koos, L. Rusko, E. Valeriano, B. Swaminathan, L. Wu, P. Bhatia, T. Kass-Hout, and E. Bas, "Decipher-MR: A Vision-Language Foundation Model for 3D MRI Representations." *npj Digit. Med.*, vol. 9, art. 419, 2026. [Paper]

[j 2] **X. Xu**, M. Klasky, M. McCann, J. Hu, and J. A. Fessler, "Swap-Net: A Memory-Efficient 2.5D Network for Sparse-View 3D Cone Beam CT Reconstruction to ICF Applications." *IEEE Trans. Comput. Imaging*, vol. 11, pp. 872–887, 2025. [Paper]

[j 3] **X. Xu***, W. Gan, S. V. V. N. Kothapalli, D. A. Yablonskiy, and U. S. Kamilov, "CoRRECT: A Deep Unfolding Framework for Motion-Corrected Quantitative R2* Mapping." *J. Math. Imaging Vis.*, vol. 67, no. 2, art. 20, 2025. [Paper]

[j 4] Z. Li, Y. Jia, **X. Xu**, J. Hu, J. A. Fessler, and Y. K. Dewaraja, "Shorter SPECT Scans Using Self-Supervised Coordinate Learning to Synthesize Skipped Projection Views." *EJNMMI Phys.*, vol. 12, art. 47, 2025. [Paper]

[j 5] T. Hong, **X. Xu**, J. Hu, and J. A. Fessler, "Provable Preconditioned Plug-and-Play Approach for Compressed Sensing MRI Reconstruction." *IEEE Trans. Comput. Imaging*, 2024. [Paper]

[j 6] D. Serino, B. Nadiga, M. Klasky, **X. Xu**, and J. A. Fessler, "Reconstructing Richtmyer–Meshkov Instabilities from Noisy Radiographs Using Low-Dimensional Features and Attention-Based Neural Networks." *Opt. Express*, 2024. [Paper]

[j 7] Z. Li, **X. Xu**, J. Hu, J. A. Fessler, and Y. Dewaraja, "Reducing SPECT Acquisition Time by Predicting Missing Projections with Single-Scan Self-Supervised Coordinate-Based Learning." *J. Nucl. Med.*, vol. 64, suppl. 1, art. P1014, 2023. [Paper]

[j 8] A. Benfenati, P. Cascarano, U. S. Kamilov, and **X. Xu**, "Constrained Regularization by Denoising with Automatic Parameter Selection." *IEEE Signal Process. Lett.*, 2023. [Paper]

[j 9] Z. Li*, J. Hu*, **X. Xu**, L. Shen, and J. A. Fessler, "Accelerated Wirtinger Flow with Score-Based Image Priors for Holographic Phase Retrieval in Poisson-Gaussian Noise Conditions." *IEEE Trans. Comput. Imaging*, 2024. [Paper]

[j 10] S. Kahali, S. V. V. N. Kothapalli, **X. Xu**, U. S. Kamilov, and D. A. Yablonskiy, "Deep-Learning-Based Accelerated and Noise-Suppressed Estimation (DANSE) of Quantitative Gradient Recalled Echo (qGRE) MRI Metrics Associated with Human Brain Neuronal Structure and Hemodynamic Properties." *NMR Biomed.*, vol. 36, no. 5, art. e4883, 2023. [Paper]

[j 11] **X. Xu**, S. V. V. N. Kothapalli, S. Kahali, U. S. Kamilov, and D. A. Yablonskiy, "Learning-Based Motion Artifact Removal Networks for Quantitative R2* Mapping." *Magn. Reson. Med.*, vol. 88, no. 1, pp. 106–119, 2022. [Paper]

[j 12] **X. Xu**, Y. Sun, J. Liu, B. Wohlberg, and U. S. Kamilov, "Provable Convergence of Plug-and-Play Priors with MMSE Denoisers." *IEEE Signal Process. Lett.*, vol. 27, pp. 1280–1284, 2020. [Paper]

[j 13] **X. Xu***, Y. Sun*, Z. Wu*, B. Wohlberg, and U. S. Kamilov, "Scalable Plug-and-Play ADMM with Convergence Guarantees." *IEEE Trans. Comput. Imaging*, vol. 7, pp. 849–863, 2021. [Paper]

[j 14] J. Liu, Y. Sun, W. Gan, **X. Xu**, B. Wohlberg, and U. S. Kamilov, "SGD-Net: Efficient Model-Based Deep Learning with Theoretical Guarantees." *IEEE Trans. Comput. Imaging*, vol. 7, pp. 598–610, 2021. [Paper]

Conferences *(* indicates equal contribution)*

[c 1] L. Da, R. Wang, **X. Xu**, P. Bhatia, T. Kass-Hout, H. Wei, and C. Xiao, "FlanS: A Foundation Model for Free-Form Language-Based Segmentation in Medical Images." *Proc. ACM SIGKDD Conf. Knowledge Discovery and Data Mining (KDD)*, pp. 404–414, 2025. [Paper] (**Acceptance rate: 19%**).

[c 2] J. Hu, B. Song, **X. Xu**, L. Shen, and J. A. Fessler, "Learning Image Priors through Patch-Based Diffusion Models for Solving Inverse Problems." *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. [Paper] (**Acceptance rate: 26.07%**).

[c 3] Z. Li*, J. Hu*, **X. Xu**, L. Shen, and J. A. Fessler, "Poisson-Gaussian Holographic Phase Retrieval with Score-Based Image Prior." *NeurIPS Workshop*, 2023. [Paper]

[c 4] **X. Xu**, J. A. Fessler, M. Klasky, G. S. Sidharth, J. L. Schei, and M. T. McCann, "An End-to-End Learning Approach for Subpixel Feature Extraction." *Optica Imaging Congress*, Technical Digest Series, paper JW2A.1, 2023. [Paper]

[c 5] **X. Xu***, J. Liu*, W. Gan, S. Shoushtari, and U. S. Kamilov, "Online Deep Equilibrium Learning for Regularization by Denoising." *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, pp. 25363–25376, 2022. [Paper] (**Acceptance rate: 25.6%**).

[c 6] Y. Hu, J. Liu, **X. Xu**, and U. S. Kamilov, "Monotonically Convergent Regularization by Denoising." *Proc. IEEE Int. Conf. Image Processing (ICIP)*, pp. 426–430, 2022. [Paper]

[c 7] **X. Xu***, A. H. Al-Shabili*, I. Selesnick, and U. S. Kamilov, "Bregman Plug-and-Play Priors." *Proc. IEEE Int. Conf. Image Processing (ICIP)*, pp. 241–245, 2022. [Paper]

[c 8] J. Liu, Y. Sun, W. Gan, **X. Xu**, B. Wohlberg, and U. S. Kamilov, "Stochastic Deep Unfolding for Imaging Inverse Problems." *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1395–1399, 2021. [Paper]

[c 9] **X. Xu**, J. Liu, Y. Sun, B. Wohlberg, and U. S. Kamilov, "Boosting the Performance of Plug-and-Play Priors via Denoiser Scaling." *Proc. Asilomar Conf. Signals, Systems, and Computers (ACSSC)*, pp. 1305–1312, 2020. [Paper] (**Oral**).

[c 10] **X. Xu**, O. Dhifallah, H. Mansour, P. T. Boufounos, and P. V. Orlik, "Robust 3D Tomographic Imaging of the Ionospheric Electron Density." *Proc. IEEE Int. Geoscience and Remote Sensing Symposium*

(IGARSS), pp. 437–440, 2020. [Paper] (**Oral**).

[c 11] J. Liu, Y. Sun, **X. Xu**, and U. S. Kamilov, "Image Restoration Using Total Variation Regularized Deep Image Prior." *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, pp. 7715–7719, 2019. [Paper]

[c 12] **X. Xu** and U. S. Kamilov, "SignProx: One-Bit Proximal Algorithm for Nonconvex Stochastic Optimization." *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, pp. 7800–7804, 2019. [Paper] (**Oral**).

Student mentoring

Interns *07/2024 – Present*

- Chicago Park, "Next-Dense-Stride Prediction for Multimodal Autoregressive Visual Modeling"
- Xiangrui Liu, "Refined Consistency for Report Generation"
- Longchao Da, "Foundation Models for Free-Form Language-Based Segmentation"

Graduates *07/2018 – 05/2024*

- Ziyun Li, "Coordinate-based internal learning for medical imaging"
- Jason Hu, "Phase retrieval with score-based image priors"
- Yixuan Luo, "Deep-learning-based image segmentation"
- Zhixin Sun, "Neural representation for image reconstruction"
- Yongcheng Song, "Focal-plane regression for autofocus"
- Weijie Gan, "Fast MRI reconstruction and artifact correction"
- Jiaming Liu, "Total variation regularized deep image prior"
- Jiarui Xing, "Deep-learning-based image artifact correction"
- Ryogo Suzuki, "Unfolding networks for image restoration"
- Yukun Li, "Single-image denoising"
- Shiqi Xu, "Sparse Fourier ptychographic microscopy"
- Fa Long, "Dictionary learning for image restoration"

Undergraduates *07/2018 – 05/2022*

- Eddie Chandler, "Inhomogeneity correction for MRI"
- Julia Zeng, "Learning-based 3D image denoising"
- Michael Kincheloe, "Reinforcement learning for MRI artifact correction"
- Mingyang Xie, "Accelerated Newton method for CT reconstruction"
- Hao Tang, "Adversarially robust classifiers for image reconstruction"
- Gustavo Gratacós, "Image restoration with patch-wise sparse learning"
- Jhoan S. Hernandez, "Convolutional sparse learning"